MFGS2Net: A Morphological Features Guided Scale Separation Network for Remote Sensing Panoptic Segmentation

Bo Sun, Yiming Yan, *Member, IEEE*, Han Fu, Guangjun He, Yanming He, Zhengning Zhang, and Fengjiao Gao

***Abstract*— Remote sensing panoptic segmentation (RSPS) plays a crucial role in the intelligent interpretation of remote sensing data. Specifically, panoptic segmentation of airport scenes presents significant challenges due to the diverse shapes of terminal buildings, along with variations in their appearance, size, scale, materials, and color, all of which make the segmentation task highly complex. The intricate geometric shapes and varying scale parameters lead to the entangled object scales in the feature space, posing challenges for accurate segmentation. These factors collectively contribute to the complexity of panoptic segmentation for irregularly shaped buildings. Current mainstream mask-based panoptic segmentation methods do not fully consider the importance of enhancing and integrating low-level and high-level semantic features. They lack the ability to decouple features, leading to misclassification and false negatives. To solve the above-mentioned problem of panoptic segmentation of irregularly shaped buildings, this study designs a morphology-guided feature aggregation (MGFA) module to establish the correlation between low and high level semantic features at different stages. At the same time, a multiorder semantic decoupling (MOSD) module is designed to capture structural information at different scales and improve the network's understanding of complex structural information. In addition, due to the lack of datasets for transportation hub scenarios in the RSPS task, we have collected the airport remote sensing panoptic segmentation (AP-RSPS) dataset to promote the practical application of RSPS tasks. Extensive experiments on the AP-RSPS dataset and BSB aerial dataset show that the proposed morphological features guided scale separation network (MFGS2Net) outperforms several SOTA methods.**

***Index Terms*— Attention guidance, irregularly shaped building, panoptic segmentation, remote sensing images.**

Received 23 January 2025; revised 21 March 2025 and 12 April 2025; accepted 20 April 2025. Date of publication 24 April 2025; date of current version 5 May 2025. This work was supported in part by the Key Research and Development Program of Heilongjiang Province under Grant 2024ZX05B11, in part by the Central Government-Guided Special Fund for Local Science and Technology Development under Grant ZY24QY02, in part by Heilongjiang Provincial Natural Science Foundation of China under Grant PL2024F038, and in part by the National Nature Science Foundation of China under Grant 42201421 and Grant 62071136. *(Corresponding author: Yiming Yan.)*

Bo Sun and Yiming Yan are with the Key Laboratory of Advanced Marine Communication and Information Technology, Ministry of Industry and Information Technology, College of Information and Communication Engineering, Harbin Engineering University, Harbin 150009, China (e-mail: sunbo7077@hrbeu.edu.cn; yanyiming@hrbeu.edu.cn).

Han Fu and Guangjun He are with Beijing Institute of Satellite Information Engineering, State Key Laboratory of Space-Earth Integrated Information Technology, Beijing 100095, China (e-mail: fuhan2017@radi.ac.cn; hgjun2006@163.com).

Yanming He and Zhengning Zhang are with Space Star Technology Company Ltd., Beijing 100095, China (e-mail: 1335770475@hrbeu.edu.cn; 23880666@qq.com).

Fengjiao Gao is with the Intelligent Manufacturing Research Institute, Heilongjiang Academy of Sciences, Harbin 150001, China (e-mail: gaofengjiao@haai.org.cn).

Digital Object Identifier 10.1109/TGRS.2025.3563917

I. Introduction

REMOTE sensing panoptic segmentation (RSPS) is an advanced image segmentation task that combines the characteristics of semantic segmentation [1], [2], [3] and instance segmentation [4], [5], [6] simultaneously performing category classification of both foreground and background, as well as distinguishing between instances of the same category. Compared to methods such as object detection [7], [8], land cover semantic segmentation, instance segmentation, and classification [9], [10], [11] which focus on interpreting specific regions of interest within an image, RSPS provides a comprehensive understanding and global cognition of complex scenes. However, research in RSPS remains relatively limited. The complexity of remote sensing images, characterized by diverse land cover types, varying object scales, and irregular shapes, further complicates the application of RSPS. The need for more robust models capable of addressing these unique challenges is urgent for advancing remote sensing analysis.

RSPS for airport transportation hub scenes plays a crucial role in the intelligent interpretation and application of remote sensing imagery, with significant practical value in civil airport management [12] and military battlefield planning. Buildings in typical residential areas often exhibit more regular and predictable shapes. These structures are usually arranged in a grid-like fashion with consistent layouts, where the forms are less complex and the spatial relationships between buildings are more uniform, as shown in Fig. 1(a)–(c). This predictable arrangement and regularity in building shapes and orientations make the RSPS significantly easier. However, in airports, remote sensing images exhibits diverse landform characteristics. Terminal buildings, in particular, display a variety of shapes in their horizontal layouts, including typical "linear," "satellite concourse" forms, and complex composite shapes. These forms differ significantly in appearance, color distribution, texture details, and scale [13]. As shown in Fig. 1(d), the linear terminal has narrow structure that carries rich local details, while its length provides global

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

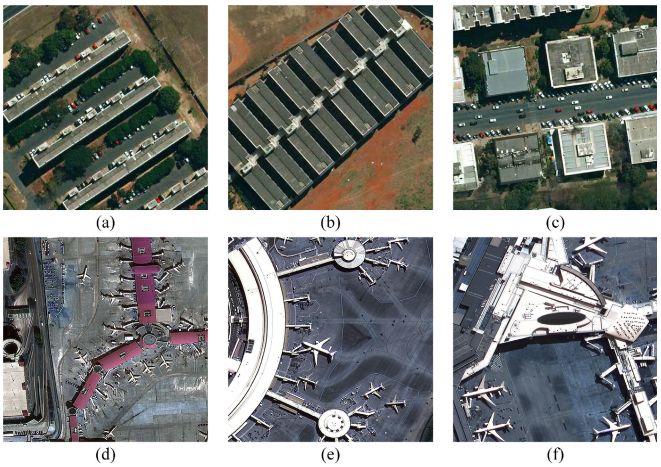


Fig. 1. Examples of the buildings with distinct characteristics. First row (a)–(c) depicts buildings with regular shapes and orderly spatial layouts, while the second row (d)–(f) showcases irregularly shaped, nonstandard airport terminals.

characteristics [14], increasing the difficulty of RSPS. The satellite concourse terminal Fig. 1(e) has a multinode, large-span spatial layout, requiring RSPS algorithms with strong global cognitive abilities to accurately identify and segment each satellite concourse. In the context of complex variations in object shapes and scale differences, an effective adaptive feature extraction strategy can enhance the model's ability to recognize diverse object shapes, while efficient multiscale feature extraction improves the model's perception and analysis capabilities across different scales. Thus, how to adaptively extract collective features for different object shapes and efficiently extract multiscale features has become a critical challenge in RSPS.

Existing panoptic segmentation methods have made positive strides in multiscale feature extraction and aggregation. JSIS-Net [15] combines instance and semantic segmentation in a single network, using the instance segmentation branch's foreground categories in postprocessing to produce the final panoptic result. IDNet [16] divided the panoptic segmentation task into independent steps for multiscale feature extraction of category information and location information, followed by information aggregation, which significantly improved feature utilization efficiency. BGRNet [17] introduces graph structures in both instance and semantic segmentation to explore the relationships between foreground objects and background. The bidirectional graph connection module efficiently propagates information across branches, leveraging their complementarity for flexible reasoning. Mask2Former [18] introduces a pixel decoder module with a multiscale deformable attention mechanism to integrate multiscale features, enhancing cross-scale information exchange and context modeling. These methods directly integrate low-level features with high-level features in the decoder, overlooking the varying complexities of different input images and the distinct requirements for boundary information in low-level features. This direct fusion of multiscale features may lead to information redundancy or cause the loss of crucial details. However, there remains a notable gap in the development of panoptic segmentation networks specifically designed to address the unique characteristics and challenges inherent in remote sensing data.

To address the challenges and limitations in RSPS, we propose a morphological features guided scale separation network (MFGS2Net) to address the challenges of irregularly shaped buildings and varying scales. Specifically, the network consists of two key modules: morphology-guided feature aggregation (MGFA) module and the multiorder semantic decoupling (MOSD) module. The MGFA module constructs an importance feature map between the backbone network features and decoder features at different stages, serving as auxiliary information to guide feature aggregation based on feature importance. This ensures the establishment of strong correlations between low-level and high-level semantic features across different stages. Additionally, to address the morphological and scale differences between various terminal buildings and structures in airport remote sensing images, the MOSD module explores the interactions between structural features at low, medium, and high scales, enhancing the network's ability to perceive and interpret complex structures and fine details within the image.

Furthermore, the lack of publicly available datasets for RSPS, particularly for critical airport transportation hub scenarios, poses a significant challenge in the interpretation of intelligent remote sensing data. To address this, we have collected and created a comprehensive airport remote sensing panoptic segmentation (AP-RSPS) dataset. This dataset presents challenges including intraclass scale variation, diverse building shapes, inconsistent lighting conditions, and occlusions from surrounding objects. By addressing the data gap for panoptic segmentation tasks, this dataset provides additional challenges and promotes further advancements in the application of panoptic segmentation. The main contributions of this article are listed as follows.

1) The MFGS2Net is introduced as a scale-separation panoptic segmentation network for RSPS, specifically designed to address the challenges posed by irregularly shaped buildings and scale variations.
2) The MGFA is proposed to address the challenges posed by the varying shape contours between buildings, which guides the weighted fusion of features and establishes effective correlations between low-level and high-level semantic information across different stages.
3) The MOSD module is introduced, which focuses on decoupling multiscale semantic features to effectively handle scale differences between buildings.
4) A comprehensive airport RSPS (AP-RSPS) dataset has been collected, which contains more challenging and diverse scenarios, enriching research in the practical applications of panoptic segmentation. Extensive experiments demonstrate that the proposed MFGS2Net outperforms current state-of-the-art methods on the AP-RSPS dataset.

II. Related Works

Currently, there are two main types of panoptic segmentation methods: two-stage panoptic segmentation and single-stage unified panoptic segmentation [19]. This section provides a brief review of relevant literature, focusing on

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

methods for the extraction and fusion of multilayer features, as well as techniques for feature enhancement.

A. Feature Extraction and Fusion in Panoptic Segmentation

Panoptic segmentation methods based on deep learning rely on the backbone to obtain high-dimensional features. Early two-stage panoptic segmentation models adopted a two-stage segmentation framework, typically consisting of three key components: a semantic segmentation branch, an instance segmentation branch, and a postprocessing module [20]. These models primarily focused on the iterative refinement of feature extraction and fusion techniques, with the goal of improving the integration of semantic and instance-level information to achieve more accurate and coherent segmentation outcomes.

Multiscale features are essential for more effective representation of image characteristics. Panoptic FPN [21] utilizes a pyramid-based feature backbone to extract multiscale features and employs a dual-branch framework to combine high-resolution low-level features with low-resolution high-level features, improving the model's ability to capture both fine details and broader context. UPSNet [22] integrates FPN outputs with separate semantic and instance segmentation heads, enabling the extraction of multiscale features. Liu introduced OANet [23], which shares the backbone features for both tasks. Panoptic-DeepLab [24] employs a dual-decoder architecture to extract semantic and instance features, which are then fused using a majority voting strategy. This approach improves the model's ability to integrate high-level semantic information with low-level instance details, enhancing segmentation accuracy. YOSO [25] introduces a feature pyramid aggregator and a separable dynamic decoder to enhance feature extraction and fusion. The separable dynamic decoder performs multihead cross-attention through dynamic convolutions, effectively integrating multiscale features for improved efficiency and accuracy in panoptic segmentation. Fernando et al. [26] proposed a lightweight multispectral segmentation pipeline that employs cross-modal attention fusion and online knowledge distillation, achieving superior performance on farmland datasets. Carvalho et al. [27] introduced a hybrid approach combining semantic segmentation and nonlearning methods, though its performance gains were limited.

In summary, these approaches fail to fully leverage the guiding and complementary roles of low-level features in high-level feature processing, as they directly integrate features across different levels. Consequently, the network is unable to adaptively aggregate features according to the specific characteristics of the image, which hampers its ability to handle the diverse shapes and complex contours of objects in remote sensing scenes, ultimately resulting in suboptimal panoptic segmentation performance.

B. Feature Enhancement in Panoptic Segmentation

Since the DETR [28] and MaX-DeepLab [29], panoptic segmentation methods have shifted toward a new end-to-end paradigm. This paradigm employs a mask-based segmentation strategy, treating both foreground objects and background elements as instances of the same target and eliminates the need for merging operators. Max-DeepLab directly predicted $\{y_i\}_{i=1}^{N} = \{(m_i, p_i(c))\}_{i=1}^{N}$, where N represents the size of set in the prediction, m_i is the mask of the prediction, and $p_i(c)$ is the probability of assigning the class c to the mask m_i. Possible classes c include thing class, stuff class, and empty class. By directly optimizing for panoptic segmentation within a unified framework, these methods emphasize enhancing multiscale features, which is crucial for achieving accurate panoptic segmentation in complex scenes.

PanopticFCN [30] enhances feature representation by using thing and stuff classes as kernels and merging kernel weights across stages. Similarly, Panoptic SegFormer [31] jointly models both foreground and background through a unified mask prediction process. Inspired by mask classification, MaskFormer and Mask2Former [18], [32] incorporates multiscale high-resolution features and a mask attention mechanism, improving performance and training efficiency, especially for small object segmentation. MP-Former [33] addresses inconsistent mask predictions by enhancing the mask attention mechanism with noisy real masks, mitigating the impact of inaccurate predictions. CMT-DeepLab [34] treated object queries as cluster centers and applied clustering to assign pixels based on feature affinity, enhancing feature extraction through an iterative process. Later, kMaX-DeepLab [35] reinterpreted cross-attention learning as clustering, which improved segmentation performance. Mask DINO [36] extended mask segmentation with the DINO object detection model integrating multiscale features and pixel embeddings. To tackle spectral variation and similarities between land cover types, RS DINO [37] incorporate a dual-channel attention mechanism. At last, Jain et al. [38] proposed OneFormer, using task-conditioned training and query-text contrastive loss to improve multitask image segmentation by enhancing feature learning across tasks.

Although traditional methods utilize attention mechanisms to capture relationships between individual semantic features and enhance feature representation, they often neglect the critical aspect of scale separation. This oversight leads to the entanglement of semantic information across different scales, undermining the accuracy of semantic interpretation. Furthermore, research in RSPS remains limited, emphasizing the need for further advancements and a more comprehensive exploration of this domain.

III. Method

A. Overall Architecture of MFGS2Net

To address the issue of black hole-like misclassification caused by the complexity of irregular building structures and ambiguous boundaries in airport environments, this article proposes the MGFA module and MOSD module. The overall architecture of the MFGS2Net network is illustrated in Fig. 2. Specifically, remote sensing images are processed through ResNet-50 [39] as the backbone encoder network, which extracts and processes the initial features. The MGFA module then establishes feature importance maps between low-level and high-level semantic features at different stages, which serve as auxiliary information to guide feature aggregation based on feature importance. This process effectively strengthens the association between features and enhances the representation of key features.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

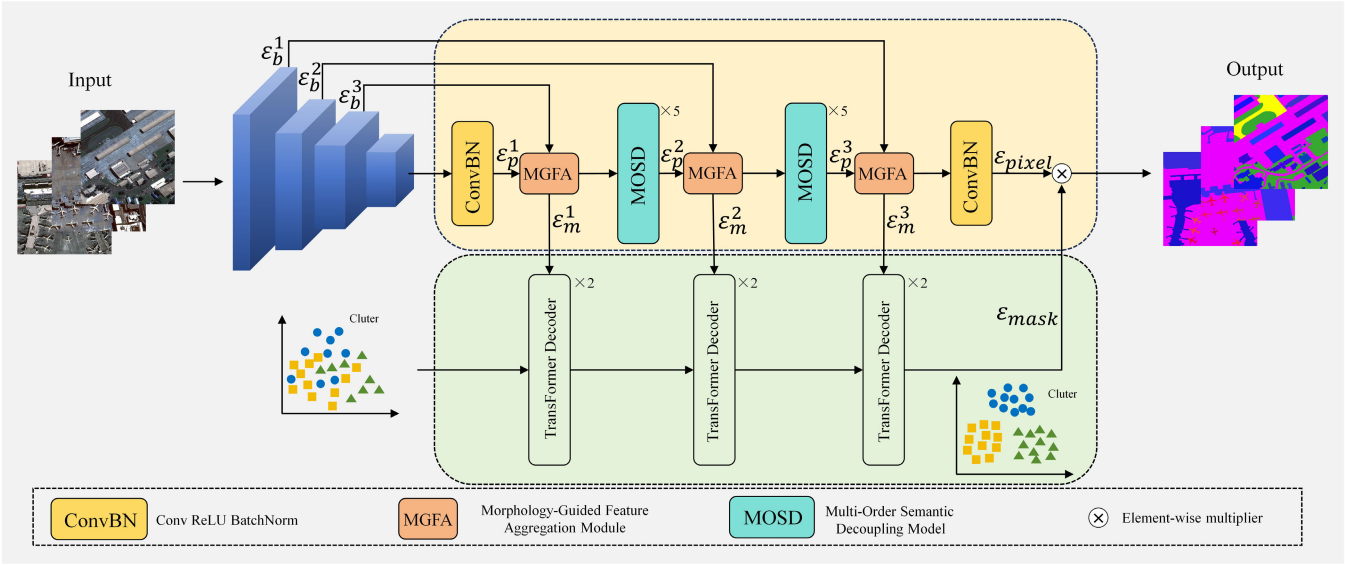


Fig. 2. Proposed MFGS2Net framework consists of three main components. First, features from ResNet50 are passed to the MGFA, which adaptively aggregates low-level and high-level features. The MOSD module then decouples and separates features across multiple scales to capture structural variations. Finally, the clustering transformer decoder refines the segmentation by leveraging learned relationships between object classes and spatial contexts, enhancing segmentation accuracy and consistency.

In addition, the MOSD module is designed to capture and parse structural information at multiple scales, integrating feature representations from different scales to improve the network's understanding of complex structural information, thereby obtaining refined pixel-level features. The extracted features, after being enhanced by multiscale and multistage information fusion branches, are input into the clustering decoder branch, where global context information is captured, further enhancing the global expressiveness of the features. This results in both mask features and clustering classification outputs. Finally, the pixel features are multiplied by the mask features and passed through a classifier to obtain the final panoptic segmentation result. Sections III-B and III-C will provide a detailed description of the working principles and design details of the MGFA and MOSD modules.

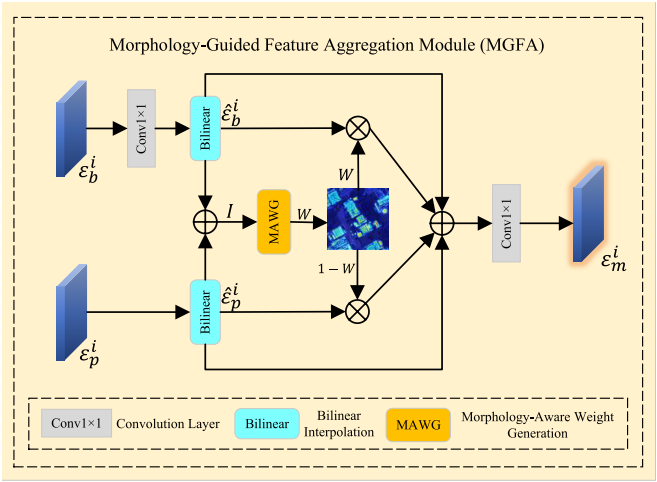


Fig. 3. MGFA consists of MAWG for weight computation and feature-weighted fusion.

B. MGFA Module

The MGFA facilitates the aggregation of low-level and high-level features through learned weights, enhancing the model's ability to capture and interpret architectural morphological features. This approach effectively addresses the shape and contour variations inherent in irregularly shaped buildings. Recent studies [40], [41], [42] have demonstrated the effectiveness of fusing features from the encoder and decoder in segmentation and other vision tasks. While low-level features are crucial for capturing fine details and recognizing object boundaries, their influence diminishes through multiple layers of the network [43]. Feature fusion strengthens information flow from shallow to deep layers, improving feature retention and gradient backpropagation. The mismatch between shallow and deep features, caused by differences in receptive fields, cannot be effectively resolved through simple addition, concatenation, or mixing operations. Given the structural uniqueness of various ground objects in airport environments, such as linear and satellite terminal buildings, properly integrating and fully utilizing multilevel features from the encoder to the decoder is essential.

To address the shape and contour variations characteristic of irregularly shaped buildings, our MGFA module builds upon the dynamic fusion framework proposed in [41] and the geometric-aware fusion approach in [40]. It introduces morphology-guided weight generation to facilitate context-sensitive feature aggregation. Fig. 3 illustrates the details of the hybrid fusion scheme based on the proposed MGFA module. ε_b^i and ε_p^i are inputs, which are first adjusted in both channel dimension and feature size using a Conv1 $\times$ 1 and bilinear interpolation, resulting in $\hat{\varepsilon}_b^i$ and $\hat{\varepsilon}_p^i$. Specifically, ε_b^i represents the features of the backbone network, where $i \in \{1, 2, 3\}$ corresponds to the stage indices of ResNet-50. ε_p^i represents the features produced by the decoder module, where $i \in \{1, 2, 3\}$ corresponds to the stage indices of the decoder. The initial feature fusion result I is obtained by performing pixel-wise addition. The morphology feature weight W is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

then obtained through morphology-aware weight generation (MAWG), and the final output ε_m^i is generated by applying the learned weights, as shown in (1) and (2). The core of the MGFA lies in the use of the MAWG module, which computes morphology weights for feature modulation

$$\begin{cases} \hat{\varepsilon}_b^i = \text{Bilinear}\left(\text{Conv}_{1\times1}\left(\varepsilon_b^i\right)\right) \\ \hat{\varepsilon}_p^i = \text{Bilinear}\left(\varepsilon_p^i\right) \end{cases} \tag{1}$$

$$\varepsilon_m^i = \text{Conv}_{1\times1}\left(\hat{\varepsilon}_b^i \cdot W + \hat{\varepsilon}_p^i \cdot (1-W) + \hat{\varepsilon}_b^i + \hat{\varepsilon}_p^i\right). \tag{2}$$

The MAWG module is embedded in the MGFA, which aims to obtain the weight W for feature modulation. The detailed process of the MAWG module is shown in Fig. 4. Let $I \in \mathbb{R}^{C\times H\times W}$ represent the input feature that is being processed, with the objective of generating an importance feature map of the same channel as I. Feature F_{ca} is first computed using the morphological self-calibration (MSC) mechanism. MSC captures global axial context by pooling in two directions: horizontal $H_p(I)$ and vertical $V_p(I)$, as shown in (3). These operations generate two distinct axial vectors [44]. By applying broadcasted addition to these vectors, the key rectangular regions are efficiently represented as F_{axial} shown in (4). The global features after pooling implicitly assign feature importance weights to different spatial locations through the addition operation. This mechanism allows the model to focus on the most relevant parts of the image while disregarding less important areas

$$\begin{cases} H_p(I)(i,1,c) = \dfrac{1}{W}\displaystyle\sum_{j=1}^{W} I(i,j,c) & \forall i \in [1,H], \\ & c \in [1,C] \\ V_p(I)(1,j,c) = \dfrac{1}{H}\displaystyle\sum_{i=1}^{H} I(i,j,c) & \forall j \in [1,W], \\ & c \in [1,C] \end{cases} \tag{3}$$

$$F_{\text{axial}} = H_p(I) \otimes \mathbf{1}_W + \mathbf{1}_H \otimes V_p(I) \tag{4}$$

where $H_p(I)(i,1,c)$ is computed by summing across all columns j for each fixed row i and channel c. $\otimes$ represents the operation of broadcasting. $\mathbf{1}_W$ and $\mathbf{1}_H$ are broadcasting vectors of size W and H.

Then, two large kernel strip convolutions are employed, with each responsible for decoupled calibration in the horizontal and vertical directions. First, the horizontal strip convolution calibrates each row of elements to adjust the horizontal shape, bringing it into closer alignment with the foreground object. Following this, batch normalization (BN) and the nonlinear activation function ReLU are applied to enhance the expressive capacity of the feature $\hat{F}_l$. Next, the vertical strip convolution further refines the shape in the vertical direction, as shown in (5). Where $\mathcal{K}$ represents the weights of the convolution kernel, (i, j) represents the position of the pixel in the image, and c represents the channel.

This separation of directional perception enables the network to focus on different geometric structures and shapes, facilitating morphological adaptation [45], [46]. It uses elongated kernels along a single spatial dimension, which are particularly effective in capturing long-range dependencies. This property makes them a valuable complement to traditional grid convolutions [47]. The weights of the strip convolutions are learnable, enabling the model to adaptively adjust its receptive field. As shown in Fig. 5, the shape of the highlighted region is modified by two strip convolutions with different weights. During training, the model learns the optimal weights that align the convolutional kernel with the foreground object, allowing it to effectively capture the most relevant features for segmentation. This process enhances the model's ability to focus on objects of interest, improving performance in complex segmentation tasks

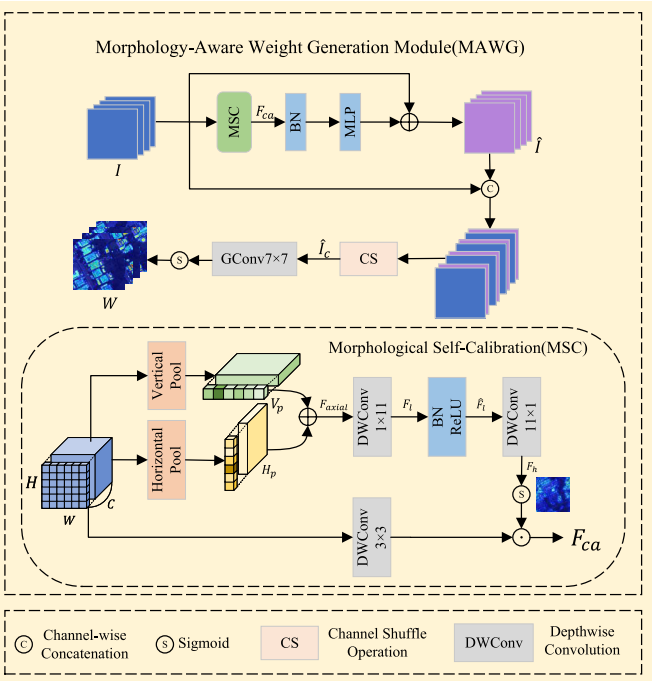


Fig. 4. MAWG module combines MSC for global context, DWConv for shape adjustment, and channel-specific weighting to prioritize important regions. Then, refines the importance feature map by generating channel-specific weights and enhancing accuracy through channel shuffle, sigmoid activation, and GConv.

$$\begin{cases} F_l = \displaystyle\sum_{t=1}^{k} F_{\text{axial}}(i, j+t-1, c)\mathcal{K}_{1\times k}(1,t,c) \\ F_h = \displaystyle\sum_{t=1}^{k} \hat{F}_l(i+t-1, j, c)\mathcal{K}_{k\times 1}(t,1,c). \end{cases} \tag{5}$$

To further capture local details within the input features, a DWConv3 × 3 is applied. The attention features, once calibrated, are integrated into the refined input features through element-wise multiplication. Inspired by Yu et al. [48], we combine MSC with the MetaNeXt structure to further improve feature representation. Specifically, BN and MLP are added after the rectangular self-calibration attention to refine features, as shown in the following equation:

$$F_{\text{ca}} = \text{DWConv}_{3\times3}(I) \odot \delta(F_h) \tag{6}$$

$$\hat{I} = \text{MLP}(\text{BN}(F_{\text{ca}})) + I. \tag{7}$$

Additionally, to generate the final refined importance feature map W, each channel of preliminary features $\hat{I}$ is adaptively

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

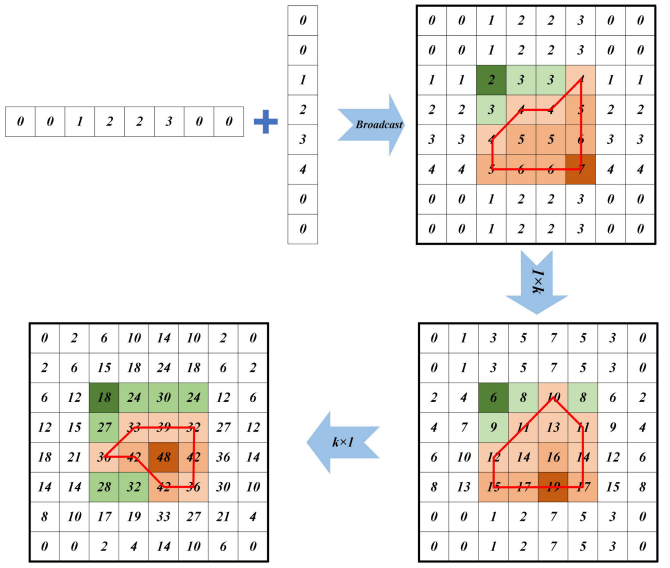


Fig. 5. Simulated how strip convolution works. Assume that the convolution kernel weights are set to 1, with zero padding applied to the outermost layer during the convolution process. Through the optimization of the weights during training, the attention region is progressively adjusted to more accurately align with the foreground object.

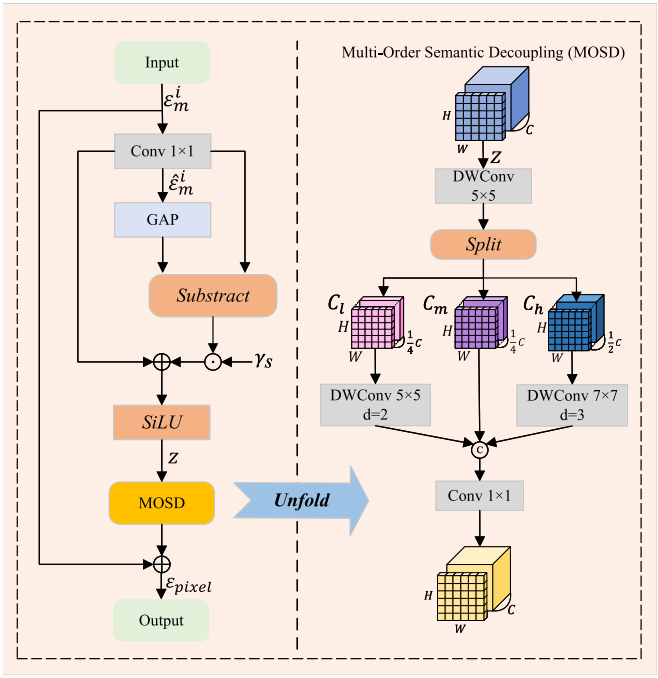


Fig. 6. Overall framework of semantic decoupling, MOSD employs three DWConv layers with varying expansions to extract multiorder features and then adaptively aggregated to enhance the model's ability to capture complex contextual relationships.

adjusted based on the corresponding input features. During this process, a channel shuffle operation [49] is applied, which alternates the rearrangement of the input feature channels

$$\hat{I}_c = \mathrm{cs}\big(\mathrm{reshape}\big(\mathrm{concact}\big(I, \hat{I}\big), (H, W, G, C/G)\big)\big). \tag{8}$$

To further improve the precision and robustness of this process, we introduce the sigmoid function δ and a group convolution (GConv) layer. The entire procedure for obtaining W can be expressed by the (8)

$$W = \delta\big(\mathrm{GConv}_{7\times7}\big(\hat{I}_c\big)\big). \tag{9}$$

C. MOSD Module

To address the issue of scale separation neglect in existing panoptic segmentation methods, we propose the MOSD module, as shown in Fig. 6. Existing methods [18], [21], [36] do not separate semantic information on different scales. This leads to missed segmentation of remote sensing images, as semantic features become entangled. The MOSD module aims to resolve this issue by ensuring proper scale separation during the semantic decoupling process. Moreover, remote sensing images, due to differing imaging mechanisms, contain far richer information than standard images, adding complexity to object recognition. Consequently, relying solely on semantic information for RSPS limits the model's ability to fully exploit feature extraction.

Both static and adaptive local perception mechanisms are integrated to extract multiscale features. Local texture features and complex global shape features are extracted using Conv1 $\times$ 1 and global average pooling (GAP), respectively, with each mechanism complementing the other. Additionally, a bias feature is computed by subtracting the complementary interaction component, which enhances local features in regions of interest while reducing redundancy. To further improve feature diversity and mitigate network biases, scale-based reweighting is applied, preventing overemphasis on irrelevant features. The process for obtaining local and global features is described in (11), where $r_s \in \mathbb{R}^{H\times W\times C}$ represents the scale factor initialized to 0

$$\hat{\varepsilon}_m^i = \mathrm{Conv}_{1\times1}\big(\varepsilon_m^i\big) \tag{10}$$

$$z = \mathrm{SiLU}\big(\hat{\varepsilon}_m^i + r_s \odot \big(\hat{\varepsilon}_m^i - \mathrm{GAP}\big(\hat{\varepsilon}_m^i\big)\big)\big). \tag{11}$$

Additionally, DWConv is employed to encode multiorder features. Unlike previous approaches that simply combine DWConv with a self-attention mechanism to model local and global interactions, three DWConv layers with expansions $(d = 1)$, $(d = 2)$, and $(d = 3)$ are utilized to simultaneously capture interactions among low, middle, and high-order features in a 1/4, 1/4, and 1/2 ratio, as depicted in Fig. 6. Specifically, for the input feature z, a DWConv5 $\times$ 5 with $(d = 1)$ is first applied to extract low-order features. The resulting output is then decomposed along the channel dimensions into $z_l \in \mathbb{R}^{C_l\times HW}$, $z_m \in \mathbb{R}^{C_m\times HW}$, and $z_h \in \mathbb{R}^{C_h\times HW}$, where $C_l + C_m + C_h = C$, as shown in the following equation:

$$(z_l, z_m, z_h) = \mathrm{Split}(\mathrm{DWConv}_{5\times5}(z)). \tag{12}$$

Next, z_m and z_h are passed through a DWConv5 $\times$ 5 with $(d = 2)$ and a DWConv7 $\times$ 7 with $(d = 3)$ to extract middle and high-order features, and z_l remains unchanged. Finally, the multiorder context is formed by concatenating the outputs of z_l, z_m, and z_h

$$\begin{cases} \hat{z}_m = \mathrm{DWConv}_{5\times5}(z_m) \\ \hat{z}_h = \mathrm{DWConv}_{7\times7}(z_h). \end{cases} \tag{13}$$

Finally, to enable adaptive aggregation of extracted features, SiLU activation functions are introduced within the branches to enhance the model's nonlinear expression, the final feature

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

$\varepsilon_{\text{pixel}}$ can be expressed as follows:

$$\varepsilon_{\text{pixel}} = \varepsilon_m^i + \text{SiLU}\left(\text{Conv}_{1\times1}\left(\text{concat}\left(z_l, \hat{z}_m, \hat{z}_h\right)\right)\right). \quad (14)$$

D. Clustering Transformer Decoder and Loss Function

Remote sensing images, due to their high information density and large size, require sophisticated processing algorithms. To address this challenge, the k-means algorithm from kMaX-DeepLab [35] is integrated into the cross-attention mechanism of the clustering transformer decoder, as shown in (15). Unlike traditional cross-attention methods, which utilize spatial-level softmax to update feature weights, this mechanism updates the weights by applying the argmax operation along the cluster dimensions. This modification effectively reduces computational complexity and eliminates the need for a multiattention mechanism. At the formula level, clustering centers group pixels with similar affinities, and the affinity logits between pixel features and clustering centers are directly mapped to the segmentation mask of softmax logits. This process leads to the generation of an accurate predicted segmentation mask

$$\begin{aligned} A &= \underset{N}{\text{argmax}}\left(C \times F^T\right) \\ \hat{C} &= A \times F \end{aligned} \quad (15)$$

where $C \in \mathbb{R}^{N\times D}$, $F \in \mathbb{R}^{HW\times D}$, $A \in \mathbb{R}^{N\times HW}$ represent the clustering centers, pixel features, and clustering assignments. Subsequently, the updated clustering centers are further aggregated using a self-attention module. After multiple updates through the transformer decoder, the clustering centers are embedded into vectors $\varepsilon_{\text{mask}}$, which are multiplied by the pixel features to generate the final prediction set.

Finally, to compute the binary mask for each segment, we take the dot product between the ith mask embedding vectors $\varepsilon_{\text{mask}}$ and the pixel-level embeddings $\varepsilon_{\text{pixel}}$. The dot product is followed by a sigmoid activation to predict the binary mask for each pixel $[h, w]$. During inference, the image is divided into multiple segments, and each pixel $[h, w]$ is assigned to one of the predicted probability-mask pairs. The pixel classification and mask assignment is done by the (17). For each pixel $[h, w]$, the final predicted class $\hat{y}[h, w]$ is determined

$$m_i[h, w] = \delta\left(\varepsilon_{\text{mask}}[:, i]^{\text{T}} \cdot \varepsilon_{\text{pixel}}[:, h, w]\right) \quad (16)$$

$$\begin{cases} c_i = \underset{c\in\{1,\ldots,K,\emptyset\}}{\text{argmax}} \; p_i(c) \\ \hat{y}[h, w] = \underset{i}{\text{argmax}}\left(p_i(c_i) \cdot m_i[h, w]\right) \end{cases} \quad (17)$$

where c_i is the most likely class label for the ith probability-mask pair. $p_i(c_i)$ is the predicted probability for class c in the ith probability-mask pair. K is the total number of classes.

To enhance the panoptic segmentation performance of MFGS2Net, a combination of multiple loss functions is utilized to guide the model's training. Following the approach [29], the model is trained using the PQ-style loss [29] alongside three auxiliary losses: instance discrimination loss $\mathcal{L}_{\text{ins}}$, mask-ID cross-entropy loss $\mathcal{L}_{\text{mask-id}}$, and semantic segmentation loss $\mathcal{L}_{\text{aux}_{\text{seg}}}$

$$t_i = -\frac{\sum_{h,w} m_{i,h,w} \cdot g_{h,w}}{\left\|\sum_{h,w} m_{i,h,w} \cdot g_{h,w}\right\|} \quad (18)$$

$$\mathcal{L}_{\text{ins}} = -\frac{\sum_{i=1}^{N} m_i \cdot \exp(t_i \cdot g/\tau)}{\sum_{i=1}^{N} \cdot \exp(t_i \cdot g/\tau)} \quad (19)$$

where $\mathcal{L}_{\text{ins}}$ is shown in (19), where g represents the pixel features, m_i denotes the downsampled ground truth mask, t_i represents the normalized feature embedding, and τ denotes the temperature. The underlying logic of the formula is that the similarity between a pixel and its corresponding class feature is computed through the dot product of their vectors. The closer the similarity, the larger the dot product, thereby reducing the loss at that pixel to be closer to zero.

As for $\mathcal{L}_{\text{mask-id}}$, $m_{i,h,w}$ represents the feature value of mask i at pixel (h, w), and the argmax operation returns $\hat{z}_{h,w}$, which represents the mask ID most relevant to that pixel. Finally, the cross-entropy loss is computed with the ground truth labels, yielding the per-pixel mask-ID cross-entropy loss

$$\hat{z}_{h,w} = \underset{i}{\text{argmax}}\, \hat{m}_{i,h,w} \quad (20)$$

$$\mathcal{L}_{\text{mask-id}} = \mathcal{L}_{\text{ce}}\left(\hat{z}_{h,w}, z_{h,w}\right). \quad (21)$$

The semantic segmentation loss is intended to help capture the semantic features of each pixel. To achieve this, the semantic head is connected to the first decoder output i_{ss} at stride 4, and the cross-entropy loss between the predicted semantic features and the ground truth m_{ss} is computed

$$\mathcal{L}_{\text{aux}_{\text{seg}}} = \mathcal{L}_{\text{ce}}(i_{\text{ss}}, m_{\text{ss}}). \quad (22)$$

The total loss function, which supervises the training of MFGS2Net, is presented as below

$$\mathcal{L} = \alpha\mathcal{L}_{\text{PQ}} + \mathcal{L}_{\text{ins}} + \beta\mathcal{L}_{\text{mask-id}} + \mathcal{L}_{\text{aux}_{\text{seg}}}. \quad (23)$$

IV. Experiment

To comprehensively validate the capabilities of MFGS2Net in handling complex geospatial scenes, we establish an evaluation framework as follows.

1) **Overall performance** comparing with state-of-the-art methods.
2) **Component effectiveness** through ablation studies.

Benchmark datasets AP-RSPS and BSB aerial are selected for their complementary characteristics: AP-RSPS emphasizes buildings with complex shapes, while BSB aerial contains diverse natural and artificial objects at varying scales. Finally, to visually demonstrate the experimental results and highlight the advantages of the proposed method, a comparison of the segmentation results with ground truth in various complex scenarios is presented.

A. Datasets Description

1) AP-RSPS: The data used in this study is the AP-RSPS dataset mentioned earlier. The dataset includes 92 typical airport images captured by the SuperView-1 satellite, with a resolution of 0.5 m. Each image comprises four spectral bands: B, G, R, and IR. The dataset encompasses ten categories, including four foreground things: airplane, ship, storage_tank, and building and six background stuff: impervious_surface, water, vegetation, bare_land, taxiway, and runway, as shown in Table I. Fig. 7 illustrates the distribution of buildings of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CATEGORY, THING/STUFF, AND NUMBER OF INSTANCES USED IN THE AP-RSPS DATASET

Category	Thing/Stuff	Polygons
Airplane	Thing	6633
Ship	Thing	2938
Storage_Tank	Thing	474
Building	Thing	35934
Imprevious_Surface	Stuff	3679
Water	Stuff	2386
Vegetation	Stuff	3477
Bare_Land	Stuff	518
Taxiway	Stuff	2008
Runway	Stuff	1493

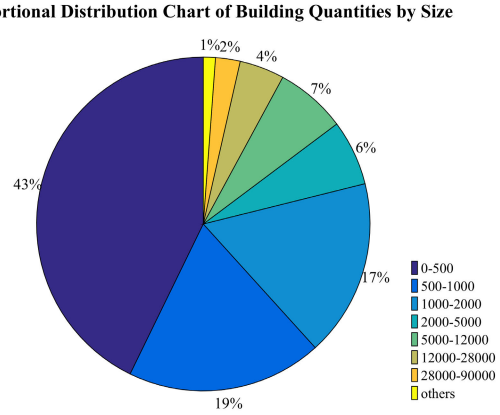


Fig. 7. Distribution of buildings of varying sizes in the AP-RSPS dataset, represented in terms of pixel count, highlights the challenges associated with segmenting buildings of different scales in remote sensing images.

varying sizes within the AP-RSPS dataset. The annotations in the AP-RSPS dataset adhere to the standard COCO format [50].

In the experiments, the images are cropped with a stride of 600 and a size of 800 × 800. After this processing, we obtain a training set containing 3929 images and a validation set containing 468 images.

2) BSB Aerial Dataset: The BSB aerial dataset [27] consists of 3000 training and 200 validation images, all captured in urban areas of Brasilia, Brazil. Each image has a resolution of 512 × 512 pixels and is densely annotated with 11 foreground things (swimming pool, harbor, vehicle, boat, sports court, soccer field, commercial building, commercial building block, residential building, house, and small construction) and three background stuff (street, permeable area, and lake) as shown in Table II. The permeable area class contains the largest number of annotated pixels, as it encompasses various types of natural soil vegetation and vehicle.

Experiments on this dataset further validate the proposed algorithm's effectiveness. The dataset's diversity in urban and natural features, along with variations in scale, texture, and context, provides an ideal benchmark for evaluating panoptic segmentation performance.

TABLE II
CATEGORY, THING/STUFF, AND NUMBER OF INSTANCES USED IN THE BSB AERIAL DATASET

Category	Thing/Stuff	Polygons
Street	Stuff	2638
Permeable_Area	Stuff	2968
Lake	Stuff	297
Swimming_Pool	Thing	4835
Harbor	Thing	121
Vehicle	Thing	84675
Boat	Thing	548
Spotrs_Court	Thing	613
Soccer_Field	Thing	89
Com_Building	Thing	3796
Res_Building	Thing	1654
Com_Building_Block	Thing	201
House	Thing	5061
Small_Construction	Thing	4552

B. Evaluation Metrics

To evaluate the performance of the AP-RSPS and BSB aerial datasets and to compare them with SOTA methods, several evaluation metrics specifically designed for panoptic segmentation tasks are utilized. The primary metric, panoptic quality (PQ) [21], is averaged across categories, providing a comprehensive assessment by combining both recognition quality (RQ) and segmentation quality (SQ). Furthermore, the boundary IoU [51] metric is employed to evaluate segmentation accuracy at object boundaries, offering additional insights into the precision of boundary delineation. The calculation of PQ and Boundary IoU is detailed as follows:

$$\mathrm{PQ} = \frac{\sum_{(p,g)\in \mathrm{TP}} \mathrm{IOU}(p,g)}{|\mathrm{TP}| + \frac{1}{2}|\mathrm{TP}| + \frac{1}{2}|\mathrm{TN}|} \tag{24}$$

$$\mathrm{PQ} = \underbrace{\frac{\sum_{(p,g)\in \mathrm{TP}}}{|\mathrm{TP}|}}_{\mathrm{SQ}} \times \underbrace{\frac{|\mathrm{TP}|}{|\mathrm{TP}| + \frac{1}{2}|\mathrm{TP}| + \frac{1}{2}|\mathrm{TN}|}}_{\mathrm{RQ}} \tag{25}$$

$$\text{Boundary IOU}(p,g) = \frac{|(g_d \cap g) \cap (p_d \cap p)|}{|(g_d \cap g) \cup (p_d \cap p)|}. \tag{26}$$

Specifically, $\mathrm{IoU}(p,g)$ denotes the intersection between the predicted segment p and the ground truth segment g, TP refers to the matched segments where $\mathrm{IoU}(p,g) > 0.5$, FP indicates false positive predictions, and FN refers to false-negative ground truth segments. The boundary regions g_d and p_d are the sets of all pixels within d pixels distance from the ground truth and prediction contours, respectively.

Furthermore, the mean intersection over union (mIoU) and frequency-weighted intersection over union (FWIoU) [52] are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

incorporated. The mIoU calculates the average IoU across all categories, while the FWIoU weights IoU by the pixel count of each category. Mean accuracy (mACC) is also introduced to provide a comprehensive assessment of the semantic SQ and precision, calculated as follows:

$$\text{mIoU} = \frac{1}{N}\frac{\sum_i n_{ii}}{t_i + \sum_j n_{ji} - n_{ii}} \tag{27}$$

$$\text{FWIoU} = \frac{1}{\sum_i t_i}\frac{\sum_i t_i n_{ii}}{t_i + \sum_j n_{ji} - n_{ii}} \tag{28}$$

$$\text{mACC} = \frac{1}{N}\frac{\sum_i n_{ii}}{t_i} \tag{29}$$

where n_{ij} denotes the number of pixels of class i predicted to class j, N represents the total number of classes, and $t_i = \sum_j n_{ij}$ represents the total number of pixels of class i.

C. Implementation Details

The proposed method is implemented using the Detectron2 framework via PyTorch. The backbone network is employed using the ResNet-50 model pretrained on ImageNet. The method is implemented on three NVIDIA A100 GPUs with a batch size of 15 and the ADAMW optimizer. The weights for the PQ-style loss, instance discrimination loss, mask ID cross-entropy loss, and semantic segmentation loss are set to 3.0, 1.0, 0.3, and 1.0, respectively, following the parameters used in MaX-DeepLab [29].

D. Comparative Methods

In this article, two dual-stage panoptic segmentation methods (Panoptic-FPN [21], Panoptic-DeepLab [24]) and five state-of-the-art mask-based single-stage unified panoptic segmentation methods (Mask2Former [18], kMaX-DeepLab [35], Mask DINO [53], RS DINO [37], and OneFormer [38]) are selected, totaling seven baseline models. Experiments are conducted on the AP-RSPS datasets and BSB aerial datasets, with the performance of the proposed MFGS2Net compared against that of seven SOTA methods. A brief overview of these methods is provided below.

1) Panoptic-FPN: The pioneering dual-branch panoptic segmentation method, generates instance segmentation results in the instance segmentation branch, while using a shared FPN backbone in the Mask R-CNN [54] as the semantic segmentation branch. Panoptic segmentation results are obtained through postprocessing.

2) Panoptic-DeepLab: A combination of DeeplabV3+ [55] for semantic segmentation and center-point regression for instance segmentation. It represents an instance by the offset of each pixel in the instance to the centroid of the instance. The offset is minimized using mean squared error (mse) loss.

3) Mask2Former: Mask2Former uses masked attention in the transformer decoder to focus on local features, improving convergence and performance. Multiscale high-resolution features help segment small objects and regions.

4) kMaX-DeepLab: kMaX-DeepLab treats object queries as cluster centers, assigning pixels to clusters based on their feature affinity. The cluster centers and pixel features are then updated, producing denser and more segmentation-aligned cross-attention.

5) Mask DINO: Mask DINO proposes a mask branch based on multiscale features and pixel embeddings, demonstrating that detection and segmentation tasks can mutually benefit from shared architecture design. In particular, detection significantly aids the segmentation task.

6) RS DINO: Builds on Mask DINO by introducing a dual-channel attention mechanism to enhance the network's feature perception ability, addressing the "same object, different spectra, different objects, same spectra" phenomenon in remote sensing images.

7) OneFormer: OneFormer employs a task-conditioned joint training strategy to unify different GT domains (semantic, instance, or panoptic) by generating all labels from panoptic annotations, training its multitask model. It uses query-text contrastive loss during training to establish better intertask and interclass differentiation.

E. Comparisons of MFGS2Net With SOTA RSPS Models

This section presents a comprehensive comparison of the proposed MFGS2Net with several SOTA panoptic segmentation models, evaluated on two remote sensing datasets: AP-RSPS and BSB aerial. The primary goal of this experiment is to assess the model's ability to handle the unique challenges posed by remote sensing imagery, particularly its effectiveness in segmenting and recognizing objects in environments with irregular shapes and complex structures. The evaluation in Table III is conducted using several key metrics, including PQ, SQ, RQ, mIoU, and mACC. These metrics provide a detailed analysis of MFGS2Net's performance relative to existing models, highlighting its advantages in terms of both segmentation and recognition accuracy.

1) Results on AP-RSPS Dataset: The results on the AP-RSPS dataset are aimed at evaluating the model's generalization ability and performance in handling remote sensing data characterized by irregular shapes. In this context, pixel-based models such as Panoptic-FPN achieves a PQ score of 38.07, while Panoptic-DeepLab performs slightly worse with a PQ score of 36.31. Both methods struggle to strike a balance between recognition and SQ. Specifically, Panoptic-FPN suffers from a notably lower RQ score 49.41, while Panoptic-DeepLab achieves a slightly higher RQ score of 46.73.

Mask-based methods, including Mask2Former, kMaX-DeepLab, and Mask DINO, exhibit substantial improvements over the pixel-based models. Mask2Former achieves a PQ of 43.25 on the AP-RSPS dataset, with Mask DINO and RS DINO reaching even higher PQ scores of 45.29 and 45.64, respectively. However, despite these advances, their performance still lags behind that of the proposed MFGS2Net, which outperforms with a PQ of 47.64, demonstrating superior panoptic segmentation accuracy. Furthermore, the mIoU and mACC metrics further underscore MFGS2Net's superiority. MFGS2Net achieves an impressive mIoU of 60.48 and an mACC of 70.83, both of which surpass the results of other methods. These improvements in both segmentation and recognition metrics highlight MFGS2Net's

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

TABLE III
QUANTITATIVE ANALYSIS OF MODELS ON AP-RSPS AND BSB AERIAL DATASET. ALL RESULTS ARE DESCRIBED IN PERCENTAGES (%)

Method		AP-RSPS Dataset					BSB Aerial Dataset				
		PQ(%)	SQ(%)	RQ(%)	mIoU(%)	mACC(%)	PQ(%)	SQ(%)	RQ(%)	mIoU(%)	mACC(%)
Pixel-based	Panoptic-FPN [21]	38.07	68.52	49.41	50.36	58.56	52.84	76.19	63.43	66.29	75.81
	Panoptic-DeepLab [24]	36.31	69.70	46.73	53.17	65.77	61.73	86.80	69.28	**72.01**	81.87
Mask-based	Mask2Former [18]	43.25	73.72	52.87	56.89	68.60	64.75	87.00	74.06	70.54	81.38
	kMaX-DeepLab [35]	45.05	71.25	56.18	58.58	68.16	66.79	85.79	76.60	69.78	81.79
	Mask DINO [53]	45.29	74.11	55.11	52.60	63.90	67.40	**88.34**	76.16	67.81	79.94
	RS DINO [37]	45.64	73.84	55.26	53.48	64.37	66.16	88.19	74.66	68.02	79.32
	OneFormer [38]	43.27	66.57	53.13	54.76	66.58	67.60	87.59	77.06	69.45	81.75
	Ours	**47.64**	**78.73**	**59.06**	**60.48**	**70.83**	**68.47**	86.55	**78.29**	71.04	**82.39**

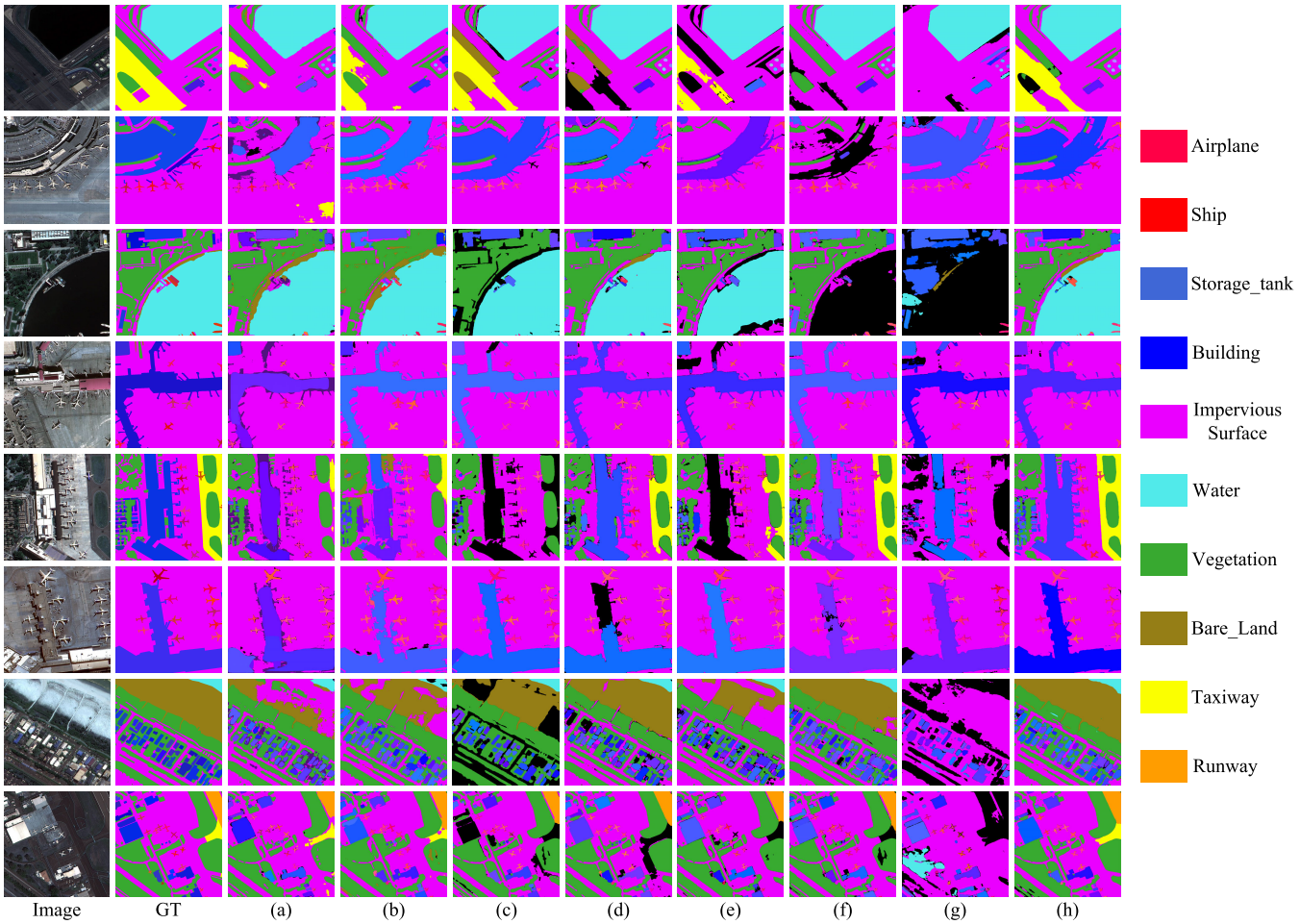


Fig. 8. Examples of panoptic segmentation results on the AP-RSPS dataset. (a) Panoptic-FPN. (b) Panoptic-DeepLab. (c) Mask2Former. (d) kMaX-DeepLab. (e) Mask DINO. (f) RS DINO. (g) OneFormer. (h) MFGS2Net.

robustness and versatility, particularly in handling remote sensing data with complex, irregular shapes, and varying scales.

The visualization results for irregularly shaped buildings on AP-RSPS dataset are shown in Fig. 8. The first column displays eight remote sensing images from the AP-RSPS dataset. The subsequent nine columns sequentially present the ground truth and predicted masks generated by comparison methods and MFGS2Net. For irregularly shaped buildings, several comparison methods fail to accurately identify the structures. Pixel-based methods such as Panoptic-FPN and Panoptic-DeepLab often misinterpret a single building as multiple distinct structures, leading to ambiguity in the segmentation results. Other mask-based methods also struggle to fully capture irregularly shaped buildings, resulting in discontinuous segmentation. In contrast, MFGS2Net significantly mitigates

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

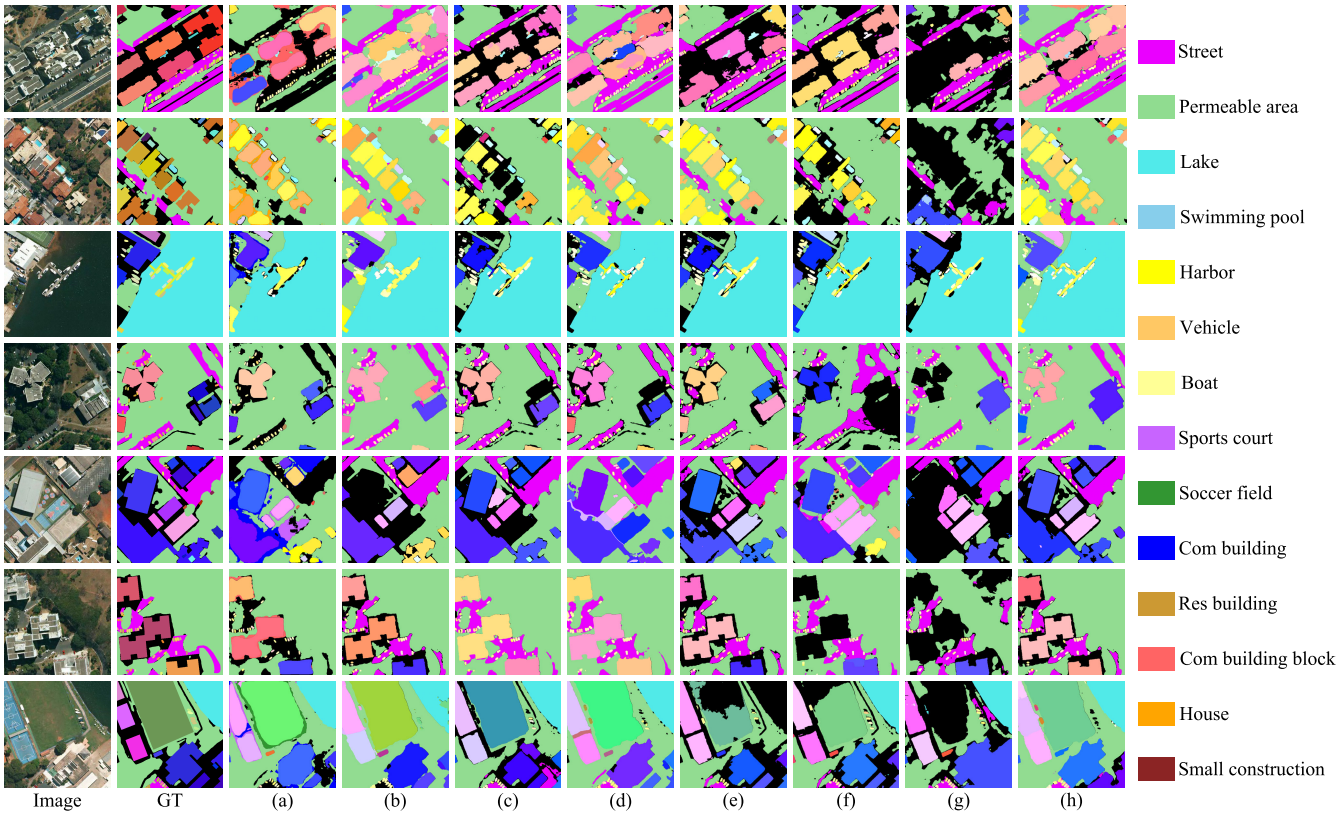


Fig. 9. Examples of panoptic segmentation results on the BSB aerial dataset. (a) Panoptic-FPN. (b) Panoptic-DeepLab. (c) Mask2Former. (d) kMaX-DeepLab. (e) Mask DINO. (f) RS DINO. (g) OneFormer. (h) MFGS2Net.

these issues by providing more accurate and complete segmentation of irregularly shaped buildings, with better delineation of their true extents, as shown in the second, fourth, fifth, and sixth rows. Furthermore, the examples in the seventh row demonstrate that MFGS2Net performs effectively for dense and small-scale ground objects.

Additionally, Fig. 10 shows the visualization of feature maps at different levels in the decoders of the MFGS2Net model on the AP-RSPS dataset, provided in the first two rows. Here, Fig. 10(a) shows the output of the last layer of features from the backbone, where the features are not well-focused. Fig. 10(b)–(d) in the figure present the visualized feature results (ε_m^1, ε_m^2, ε_m^3) after enhancement through the MOSD and MGFA modules at different stages. As the MGFA and MOSD modules progressively enhance the features, a notable increase in activations is observed, with the activation contours increasingly aligning with the target structure, which demonstrate the effectiveness of MFGS2Net at the feature level.

2) Results on BSB Aerial Dataset: The BSB aerial dataset is used to evaluate the model's ability to handle aerial images with complex background structures and highly irregular object shapes. It indicates that pixel-based models, such as Panoptic-FPN and Panoptic-DeepLab, exhibit relatively lower results across all evaluation metrics. Panoptic-FPN achieves a PQ score of 52.84, while Panoptic-DeepLab scores slightly lower at 61.73. Both models struggle to balance recognition and SQ, as reflected in their notably lower RQ scores 63.43 for Panoptic-FPN and 69.28 for Panoptic-DeepLab. Their SQ scores 76.19 for Panoptic-FPN and 86.80 for Panoptic-DeepLab are moderate, highlighting their suboptimal performance in addressing the challenges of remote sensing data with complex structures.

In contrast, mask-based methods, including Mask2Former, Kmax-DeepLab, and Mask DINO, demonstrate substantial improvements over pixel-based models in both PQ and RQ. For example, Mask2Former achieves a PQ of 64.75 on the AP-RSPS dataset, while Mask DINO and RS DINO reach even higher PQ scores of 67.40 and 66.16, respectively. These models, still face challenges in accurately segmenting complex, irregular objects and objects with varying scales. Specifically, their lower RQ scores are due to difficulties in distinguishing objects with similar appearances and intricate boundaries, and their SQ scores remain moderate. The proposed MFGS2Net outperforms all other methods with a PQ of 68.47, surpassing OneFormer, one of the top-performing models. This enhanced performance is attributed to MFGS2Net's superior capability to handle irregularly shaped objects and complex backgrounds, resulting in more precise segmentation.

Fig. 9 illustrates the prediction results of various panoptic segmentation methods presented in Table III. By comparing the complete masks and irregularly shaped buildings masks generated by each method, it becomes evident that MFGS2Net effectively minimizes segmentation errors, producing results that are most aligned with the ground truth. In the third row, other methods suffer from significant omissions and merging of small boats, whereas MFGS2Net accurately segments them. In the fifth and last rows, sports courts, soccer fields, and permeable areas, which share similar colors, are difficult for other methods to distinguish. However, MFGS2Net still performs relatively accurate segmentation in these cases. In the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
QUANTITATIVE ANALYSIS OF IOU, BOUNDARY IOU, AND FWIOU FOR THE BUILDING CLASS ON AP-RSPS DATASET

Method	IoU(%)	Boundary IoU(%)	FWIoU(%)
Panoptic-FPN [21]	64.65	37.13	71.55
Panoptic-DeepLab [24]	65.62	38.36	69.63
Mask2Former [18]	70.29	39.75	74.28
kMaX-DeepLab [35]	71.13	43.59	75.84
Mask DINO [53]	71.50	44.87	73.04
RS DINO [37]	72.34	44.59	73.52
OneFormer [38]	72.68	43.16	73.29
Ours	**76.36**	**46.27**	**78.21**

TABLE V
QUANTITATIVE ANALYSIS OF MODELS ON PARAMS, FLOPS, AND FPS ON AP-RSPS DATASET

Method	Params(M)	FLOPs(G)	FPS
Panoptic-FPN [21]	45.56	153	11.73
Panoptic-DeepLab [24]	30.27	111	7.40
Mask2Former [18]	44.26	169	8.62
kMaX-DeepLab [35]	57.44	124	5.43
Mask DINO [53]	52.09	234	4.44
RS DINO [37]	52.22	245	3.97
OneFormer [38]	64.83	199	4.34
Ours	47.09	112	5.68

remaining rows, other methods exhibit frequent classification errors, omissions, and unclear boundaries for various building types, while our method consistently provides precise segmentation.

The visualization of feature maps at different levels in the decoders of the BSB aerial dataset is provided in last two rows of Fig. 10. Ultimately, when guided by the MGFA and MOSD modules, the final output produces feature maps with well-defined boundaries and enriched information. This can be attributed to the MGFA module, which guides feature aggregation based on morphological characteristics, and the MOSD module, which decouples and integrates multiscale feature extraction to further enhance feature representation.

In summary, MFGS2Net demonstrates superior segmentation performance by effectively handling complex variations in shape and size, ensuring accurate delineation of both small and irregularly shaped structures.

3) Quantitative Results for Irregularly Shaped Buildings: To demonstrate the effectiveness of our method in handling complex, multiscale, and irregularly shaped building contours, as well as improving boundary delineation and overall scene segmentation. The results in Table IV demonstrate the superior performance of our method across three key evaluation metrics: IOU-building, BoundaryIOU-building, and FWIOU. Specifically, our approach excels in segmenting complex, multiscale, and irregularly shaped buildings contours. For IOU-building, our method achieves 76.36, significantly surpassing traditional methods such as Panoptic-FPN (64.65) and Panoptic-DeepLab (65.62). This improvement underscores the model's enhanced capability to segment buildings in urban settings with challenging shapes and scales. The MGFA is pivotal in this achievement, enabling the adaptive fusion of low- and high-level features to better capture intricate building structures. With a BoundaryIOU-building score of 46.27, our method also demonstrates substantial improvement in boundary delineation, outperforming Mask DINO (44.87). This advancement highlights its effectiveness in handling irregular and fragmented building shapes. The MOSD plays a critical role by capturing multilevel structural interactions, thereby improving boundary segmentation precision. Finally, our method achieves a FWIOU score of 78.21, outperforming all competing models, including Mask2Former (74.28) and kMaX-DeepLab (75.84). This result reflects superior performance not only in building segmentation but also in overall scene segmentation, effectively addressing both stuff and thing classes. The combined strengths of MGFA and MOSD enhance feature fusion and semantic decoupling, leading to significant improvements in SQ.

4) Model efficiency analysis: The experiment aims to evaluate the efficiency and effectiveness of the MFGS2Net model, demonstrating its lower FLOPs and superior performance across all evaluation metrics compared to baseline models. The parameter count (Params), floating-point operations (FLOPs), and frames per second (FPS) for each panoptic segmentation model on the AP-RSPS dataset are systematically presented in Table V. Notably, the MFGS2Net model has a parameter count slightly higher than earlier two-stage segmentation methods and Mask2Former, and its FLOPs are lower than all baseline models, indicating reduced model complexity. While its training efficiency is comparable to the baseline models, MFGS2Net significantly outperforms them in all performance evaluation metrics. Thus, MFGS2Net achieves an excellent balance between efficiency and effectiveness.

F. Ablation Experiment

In order to evaluate the contribution of each module in MFGS2Net, a series of ablation experiments were conducted on the AP-RSPS dataset. These experiments focused on isolating and testing individual components by removing each target module while keeping all other parameters and modules fixed. This approach provided a comprehensive understanding of how different modules influence overall performance. Additional experiments were conducted to explore the impact of various module configurations on MFGS2Net's performance. Specifically, the effect of the MGFA placement on feature fusion and model performance was investigated, and the size of DWConv in the MOSD module was assessed for its balance between computational efficiency and feature extraction. Furthermore, the influence of varying the number of MOSD modules on multiscale feature extraction was examined. These findings highlight the importance of carefully configuring the modules to optimize MFGS2Net's performance.

1) Effectiveness of each component: The first set of experiments tested the individual effectiveness of the MGFA and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

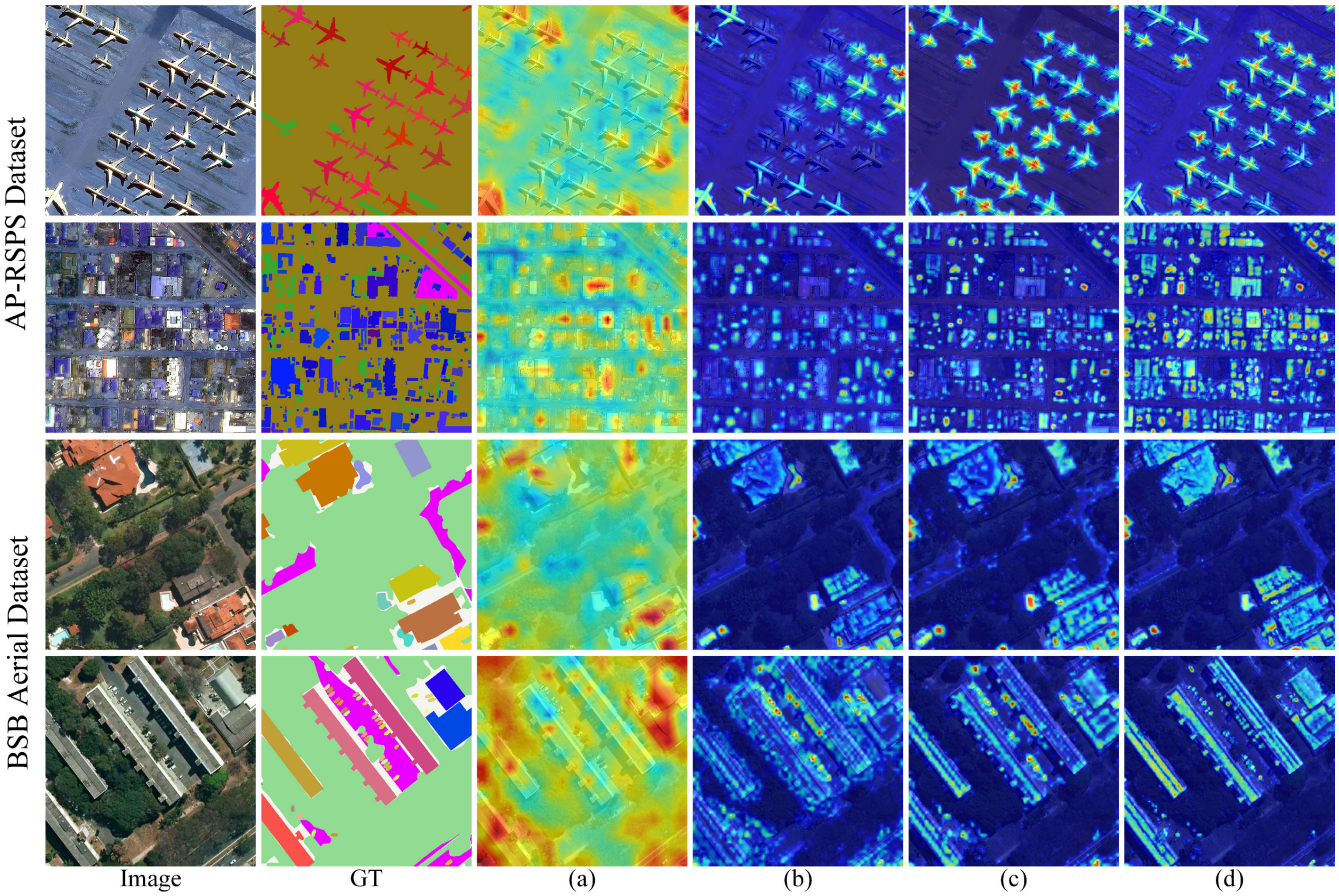


Fig. 10. Visualization of the feature maps at different levels in the decoders. (a) Feature of the last layer of the backbone network. (b) First MGFA output feature ε_m^1. (c) Second MGFA output feature ε_m^2. (d) Third MGFA output feature ε_m^3.

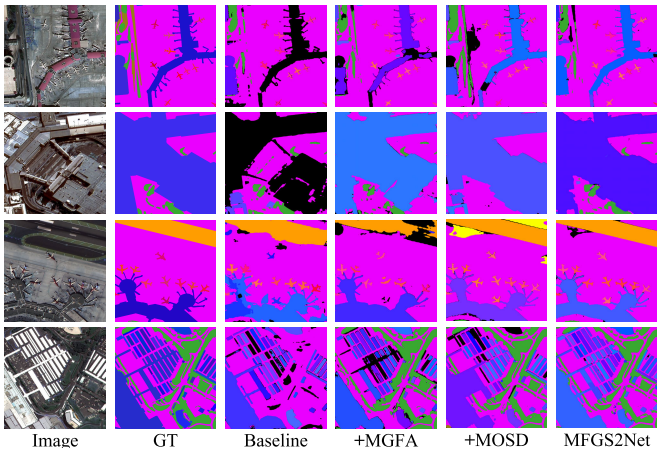


Fig. 11. Visualization of ablation experiments on the AP-RSPS datasets.

MOSD modules. By incorporating these modules, MFGS2Net showed substantial improvements in the PQ score, reflecting enhanced panoptic SQ. In this experiment, only the target module of MFGS2Net was removed, while all other parameters and modules were kept constant. This allowed for the assessment of how different modules influence the experimental outcomes. The results, presented in Table VI, clearly demonstrate the effectiveness of the MGFA and MOSD modules, as indicated by improvements in PQ, SQ, mIOU, and mACC. The inclusion of MGFA and MOSD resulted in a linear increase in PQ scores. The baseline, which lacked effective fusion of low- and high-level semantic features across layers, exhibited weak feature correlations and lower segmentation accuracy. To address this, the MGFA module was introduced, guiding the aggregation of low- and high-level features and integrating representations from multiple scales to improve the network's ability to capture complex structural information.

The baseline model failed to account for scale separation during semantic decoupling, which led to the entanglement of features across different scales. This resulted in suboptimal performance, particularly for irregularly shaped objects. To address this, a multiscale dilated depthwise convolutional semantic decoupling module was proposed, and the MOSD was integrated into the decoder to capture and analyze structural information at different scales. This improved the learning and analysis of architectural morphological features, thereby effectively handling the shape discrepancies of irregularly shaped buildings, leading to a PQ increase of 0.55. By combining the MGFA module and the MOSD module into an end-to-end learning framework, a notable performance boost was achieved on the AP-RSPS dataset. This demonstrates that the MGFA module and the MOSD module can mutually benefit each other in tackling the panoptic segmentation. The visualization results of the ablation study are presented in Fig. 11.

Furthermore, to further explore the impact of MFGS2Net on the pixel feature generation process, the feature distribution on images is visualized in Figs. 12 and 13. Specifically, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

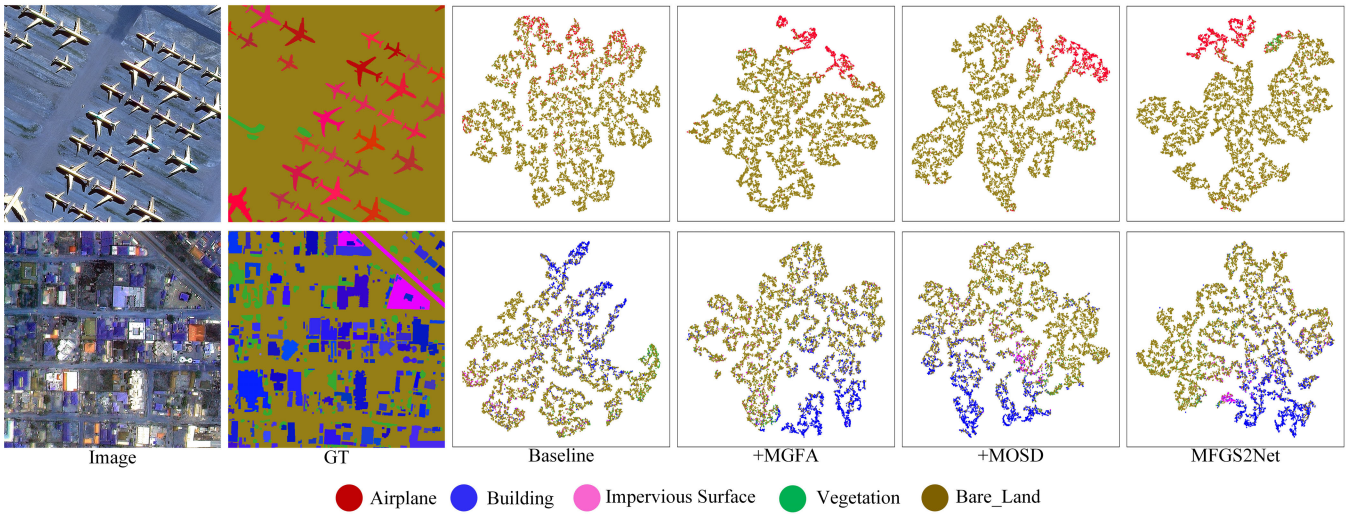


Fig. 12. Visualization of features output by the last layer of MFGS2Net. The test image is selected from AP-RSPS datasets. We implement the experiment with t-SNE.

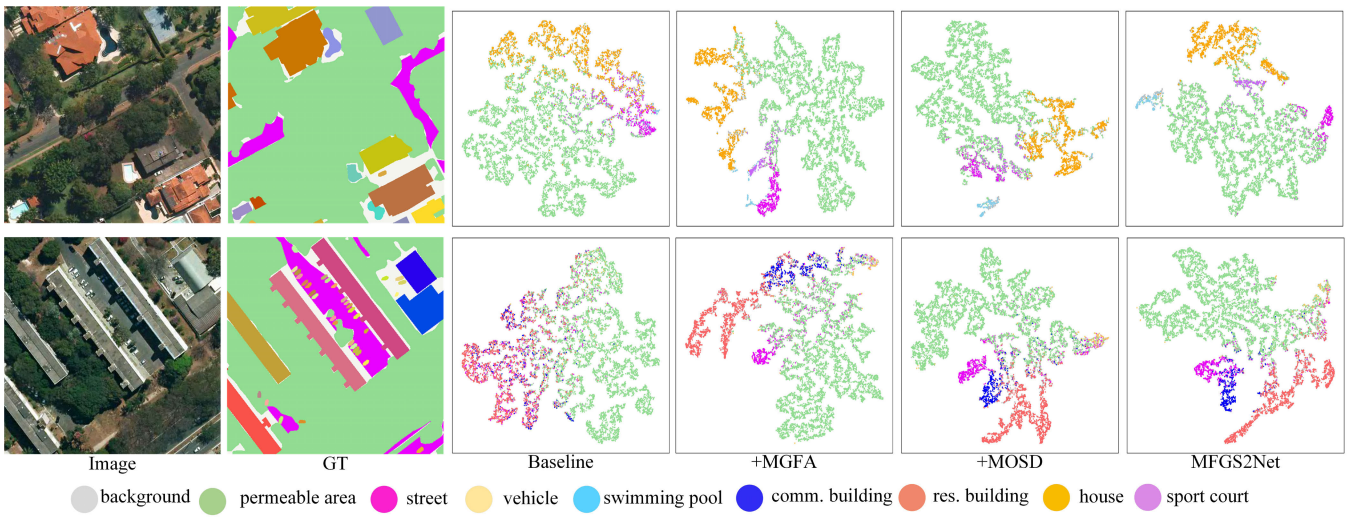


Fig. 13. t-SNE visualization of features from the last layer of MFGS2Net. The test image is selected from BSB aerial datasets.

TABLE VI
INFLUENCE OF EACH COMPONENT ON THE PERFORMANCE OF THE PROPOSED MFGS2NET

Modules		Metrics				
MGFA	MOSD	PQ(%)	SQ(%)	RQ(%)	mIOU(%)	mACC(%)
		45.05	71.25	56.18	58.58	68.16
✓		45.59	71.56	56.25	59.13	67.73
	✓	46.14	76.85	57.73	59.31	69.24
✓	✓	**47.64**	**78.73**	**59.06**	**60.48**	**70.83**

TABLE VII
INFLUENCE OF WHICH STAGE OF RESNET-50 TO PLUG THE MGFA MODULE

MGFA	PQ(%)	SQ(%)	RQ(%)	mIOU(%)	mACC(%)
Stage-1	45.22	70.90	56.35	59.51	69.72
Stage-2	45.61	70.74	57.07	59.95	69.74
Stage-3	45.25	70.84	56.49	60.13	**71.09**
Stage-1+2	45.70	71.13	56.65	59.69	69.20
Stage-1+3	46.57	71.19	57.92	60.13	69.66
Stage-2+3	46.08	70.91	57.59	60.45	70.56
Stage-1+2+3	**47.64**	**78.73**	**59.06**	**60.48**	70.83

features output by the last layer of MFGS2Net are visualized using t-SNE [56]. It is evident that the baseline methods exhibit overlap and dispersion in the t-SNE feature maps. In contrast, the proposed MFGS2Net model successfully differentiates the features. Specifically, in the second row of Fig. 12, the buildings vary in size and shape, and their colors are similar to those of the bare land, which can complicate feature separation. The feature distributions extracted by the baseline model show significant overlap between the bare land and buildings categories. At the same time, the distribution of feature points for these two categories diverges. With the guidance of the MGFA and MOSD modules, the feature points of these two categories gradually diverge, leading to a noticeable improvement in their separation and compactness.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
INFLUENCE OF DWCONV SIZE ON THE PERFORMANCE OF MFGS2NET MODEL IN MOSD

DWConv Kernel Size				Metrics		
3×3	5×5	7×7	9×9	PQ(%)	SQ(%)	RQ(%)
✓		✓		45.36	70.48	56.97
✓			✓	45.42	71.36	56.85
	✓	✓		**47.64**	**78.73**	**59.06**
	✓		✓	46.81	71.14	58.17

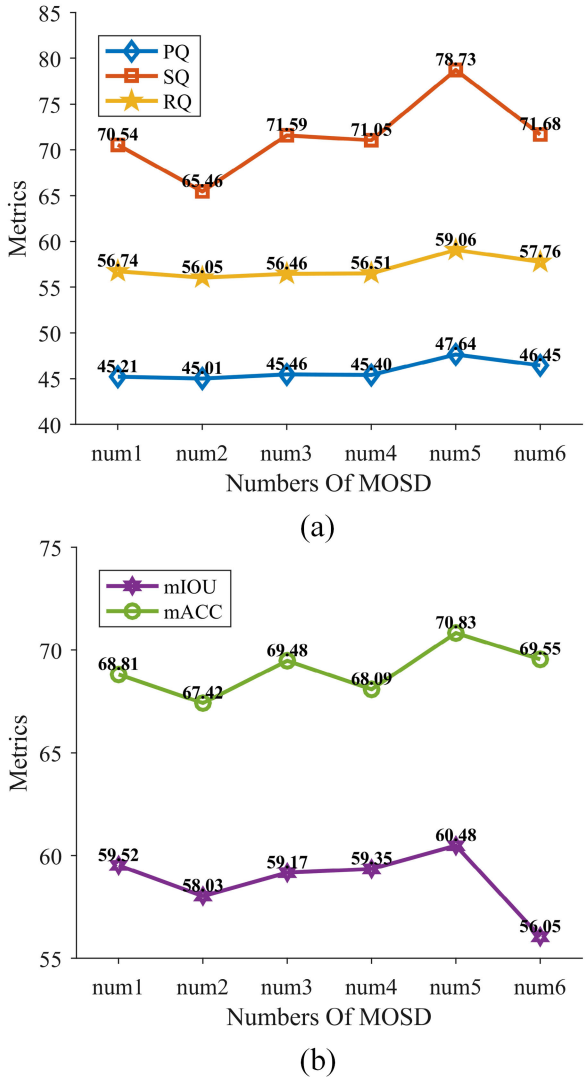


Fig. 14. Effectiveness of MOSD quantity on MFGS2Net. (a) Panoptic segmentation metrics under different numbers of MOSD. (b) General segmentation metrics under different numbers of MOSD.

In conclusion, it is evident that adding the MGFA module significantly enhances the network's ability to segment irregularly shaped buildings, with a noticeable reduction in omission errors. The inclusion of the MOSD module improves the network's capability to recognize objects of varying sizes. When both modules are incorporated simultaneously, the segmentation performance is optimized, achieving the best results.

2) Effectiveness of MGFA Location: The MGFA module plays a crucial role in linking feature representations from the backbone network to those in the decoder, while also applying morphology-aware feature weighting to guide feature aggregation. To assess the optimal placement of the MGFA module within the network architecture, we conducted an ablation study where the module was integrated at different stages of the ResNet-50 backbone (Stages 1 to 3). The experimental results, as presented in Table VII, demonstrate that when MGFA was applied to a single stage, the performance improvement was minimal. However, when MGFA was applied across multiple stages in a pairwise manner, a noticeable improvement in performance was observed. The best results were achieved when the module was incorporated after all stages of the ResNet-50 backbone, confirming the importance of using MGFA across all stages for optimal feature aggregation and network guidance.

3) Effectiveness of DWConv Size on MOSD: To explore the impact of depthwise convolution (DWConv) kernel sizes on the performance of the MOSD module, we conducted experiments using kernel sizes of 3×3, 5×5, 7×7, and 9×9. These kernels were tested individually and in combination to evaluate their effect on semantic decoupling. The results, summarized in Table VIII, reveal that when the 3×3 DWConv was combined with other sizes, the PQ metric was significantly lower, suggesting insufficient decoupling ability. Similarly, the 9×9 kernel underperformed due to the excessively large receptive field, which caused excessive information dilution. In contrast, the 5×5 and 7×7 DWConv kernels demonstrated the best performance in extracting mid-to-high-level features. This can be attributed to their ability to preserve information richness while effectively promoting the decoupling and recombination of semantic features.

4) Effectiveness of MOSD quantity on MFGS2Net: A final ablation study investigated the impact of varying the number of MOSD modules on MFGS2Net's performance. We tested configurations with 1 to 6 MOSD modules, with a focus on panoptic segmentation metrics as well as general segmentation performance. The ablation study, detailed in Fig. 14, explores configurations with 1 to 6 MOSD modules, focusing on panoptic segmentation metrics and general segmentation metric. The results reveal that using fewer modules restricts the network's ability to effectively guide feature learning. However, embedding an excessive number of such modules introduces redundant features, increasing the difficulty of model optimization and leading to performance degradation and overfitting. The optimal configuration identified as five MOSD modules. This further confirms that the selection of the module count is a crucial factor influencing the model's performance.

V. CONCLUSION

In this article, we propose MFGS2Net, a MGFA scale separation network for remote sensing image panoptic segmentation. The MGFA module learns adaptive morphological weights, guiding the aggregation of low-level and high-level features to tackle the segmentation challenges of irregularly shaped buildings in airport environments. The MOSD module

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

enhances the network's ability to interpret complex structures by exploring feature interactions across scales. Integrated within an end-to-end learning framework, MFGS2Net outperforms several state-of-the-art methods on the BSB aerial dataset and newly introduced AP-RSPS dataset.

However, the model's computational complexity increases with the integration of multiple modules, which could impact real-time performance on large-scale datasets. Future work will focus on improving efficiency and exploring the model's application to other domains and data modalities.

REFERENCES

[1] Z. Wang, S. Zhang, C. Zhang, and B. Wang, "Hidden feature-guided semantic segmentation network for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603417.

[2] H.-F. Zhong, Q. Sun, H.-M. Sun, and R.-S. Jia, "NT-Net: A semantic segmentation network for extracting lake water bodies from optical remote sensing images based on transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5627513.

[3] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, 2022.

[4] K. Chen et al., "RSPrompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701117.

[5] L. Fang, Y. Jiang, Y. Yan, J. Yue, and Y. Deng, "Hyperspectral image instance segmentation using spectral–spatial feature pyramid network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502613.

[6] M. Luo, T. Zhang, S. Wei, and S. Ji, "SAM-RSIS: Progressively adapting SAM with box prompting to remote sensing image instance segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4413814.

[7] Y. Li, Q. Hou, Z. Zheng, M.-M. Cheng, J. Yang, and X. Li, "Large selective kernel network for remote sensing object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16794–16805.

[8] D. Yu and S. Ji, "A new spatial-oriented object detection framework for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4407416.

[9] X. Li, M. Ding, and A. Pižurica, "Deep feature fusion via two-stream convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2615–2629, Apr. 2020.

[10] W. Chen, S. Ouyang, W. Tong, X. Li, X. Zheng, and L. Wang, "GCSANet: A global context spatial attention deep learning network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1150–1162, 2022.

[11] X. Li, M. Ding, Y. Gu, and A. Pižurica, "An end-to-end framework for joint denoising and classification of hyperspectral images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 7, pp. 3269–3283, Jul. 2023.

[12] J. Hou, Z. Guo, Y. Feng, Y. Wu, and W. Diao, "SPANet: Spatial adaptive convolution based content-aware network for aerial image semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 2192–2204, 2023.

[13] Z. Qin, X. Li, and Y. Gu, "Hemisphere harmonics basis: A universal approach to remote sensing BRDF approximation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[14] J. Hou, Z. Guo, Y. Wu, W. Diao, and T. Xu, "BSNet: Dynamic hybrid gradient convolution based boundary-sensitive network for remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604912.

[15] D. de Geus, P. Meletis, and G. Dubbelman, "Panoptic segmentation with a joint semantic and instance segmentation network," 2018, *arXiv:1809.02110*.

[16] G. Lin, S. Li, Y. Chen, and X. Li, "IDNet: Information decomposition network for fast panoptic segmentation," *IEEE Trans. Image Process.*, vol. 33, pp. 1487–1496, 2023.

[17] Y. Wu et al., "Bidirectional graph reasoning network for panoptic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9080–9089.

[18] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1290–1299.

[19] Y. Chuang, S. Zhang, and X. Zhao, "Deep learning-based panoptic segmentation: Recent advances and perspectives," *IET Image Process.*, vol. 17, no. 10, pp. 2807–2828, Aug. 2023.

[20] X. Li and D. Chen, "A survey on deep learning-based panoptic segmentation," *Digit. Signal Process.*, vol. 120, Jan. 2022, Art. no. 103283.

[21] A. Kirillov, K. He, R. Girshick, C. Rother, and P. Dollár, "Panoptic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 9404–9413.

[22] Y. Xiong et al., "UPSNet: A unified panoptic segmentation network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8818–8826.

[23] H. Liu et al., "An end-to-end network for panoptic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6172–6181.

[24] B. Cheng et al., "Panoptic-DeepLab: A simple, strong, and fast baseline for bottom-up panoptic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 12475–12485.

[25] J. Hu, L. Huang, T. Ren, S. Zhang, R. Ji, and L. Cao, "You only segment once: Towards real-time panoptic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 17819–17829.

[26] T. Fernando, C. Fookes, H. Gammulle, S. Denman, and S. Sridharan, "Toward on-board panoptic segmentation of multispectral satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5402312.

[27] O. L. F. de Carvalho, O. A. de Carvalho Jr., A. O. de Albuquerque, N. C. Santana, and D. L. Borges, "Rethinking panoptic segmentation in remote sensing: A hybrid approach using semantic segmentation and non-learning methods," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[28] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.

[29] H. Wang, Y. Zhu, H. Adam, A. Yuille, and L.-C. Chen, "MaX-DeepLab: End-to-end panoptic segmentation with mask transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 5463–5474.

[30] Y. Li et al., "Fully convolutional networks for panoptic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 214–223.

[31] Z. Li et al., "Panoptic SegFormer: Delving deeper into panoptic segmentation with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 1280–1289.

[32] B. Cheng, A. G. Schwing, and A. Kirillov, "Per-pixel classification is not all you need for semantic segmentation," in *Proc. NIPS*, vol. 34, Dec. 2021, pp. 17864–17875.

[33] H. Zhang et al., "MP-former: Mask-piloted transformer for image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 18074–18083.

[34] Q. Yu et al., "CMT-DeepLab: Clustering mask transformers for panoptic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 2550–2560.

[35] Q. Yu et al., "K-means mask transformer," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 288–307.

[36] H. Zhang et al., "DINO: DETR with improved DeNoising anchor boxes for end-to-end object detection," 2022, *arXiv:2203.03605*.

[37] Z. Li et al., "RS DINO: A novel panoptic segmentation algorithm for high resolution remote sensing images," in *Proc. 11th Int. Conf. Agro-Geoinformatics (Agro-Geoinformatics)*, Jul. 2023, pp. 1–5.

[38] J. Jain, J. Li, M. Chiu, A. Hassani, N. Orlov, and H. Shi, "OneFormer: One transformer to rule universal image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 2989–2998.

[39] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[40] Y. Wang, H. Zhang, Y. Hu, X. Hu, L. Chen, and S. Hu, "Geometric boundary guided feature fusion and spatial-semantic context aggregation for semantic segmentation of remote sensing images," *IEEE Trans. Image Process.*, vol. 32, pp. 6373–6385, 2023.

[41] S. Zhang, B. Zhang, Y. Wu, H. Zhou, J. Jiang, and J. Ma, "SegCLIP: Multimodal visual-language and prompt learning for high-resolution remote sensing semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5646316.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.

[42] Z. He et al., "Single image super-resolution based on progressive fusion of orientation-aware features," *Pattern Recognit.*, vol. 133, Jan. 2023, Art. no. 109038.
[43] C. Lu, X. Zhang, K. Du, H. Xu, and G. Liu, "CTCFNet: CNN-transformer complementary and fusion network for high-resolution remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5408717.
[44] Q. Hou, L. Zhang, M.-M. Cheng, and J. Feng, "Strip pooling: Rethinking spatial pooling for scene parsing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4003–4012.
[45] C. Peng, X. Zhang, G. Yu, G. Luo, and J. Sun, "Large kernel matters—Improve semantic segmentation by global convolutional network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1743–1751.
[46] X. Ding, X. Zhang, J. Han, and G. Ding, "Scaling up your kernels to 31×31: Revisiting large kernel design in CNNs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 11963–11975.
[47] M.-H. Guo, C.-Z. Lu, Q. Hou, Z. Liu, M. Cheng, and S. Hu, "SegNeXt: Rethinking convolutional attention design for semantic segmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, Jan. 2022, pp. 1140–1156.
[48] W. Yu, P. Zhou, S. Yan, and X. Wang, "InceptionNeXt: When inception meets ConvNeXt," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 5672–5683.
[49] X. Zhang, X. Zhou, M. Lin, and J. Sun, "ShuffleNet: An extremely efficient convolutional neural network for mobile devices," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6848–6856.
[50] T. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, Zurich, Switzerland. Cham, Switzerland: Springer, 2014, pp. 740–755.
[51] B. Cheng, R. Girshick, P. Dollár, A. C. Berg, and A. Kirillov, "Boundary IoU: Improving object-centric image segmentation evaluation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15334–15342.
[52] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[53] F. Li et al., "Mask DINO: Towards a unified transformer-based framework for object detection and segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3041–3050.
[54] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2961–2969.
[55] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with Atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.
[56] L. Van der Maaten and G. E. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, Jan. 2008.



Bo Sun received the B.S. degree in electronic information engineering from the School of Hainan University, Haikou, China, in 2023.

He is currently working toward the master's degree in electronic information with the School of Information and Communication Engineering, Harbin Engineering University, Harbin, China. His primary research interests include deep learning and remote sensing image segmentation.



Yiming Yan (Member, IEEE) received the B.E., M.S., and Ph.D. degrees in electronics and information engineering, and information and communication engineering from Harbin Institute of Technology, Harbin, China, in 2006, 2008, and 2013, respectively.

From 2013 to 2016, he worked as a Post-Doctoral Researcher with the Department of Control Science and Engineering, Harbin Institute of Technology, to study remote sensing image processing and 3-D reconstruction. He has been with Harbin Engineering University as an Associate Professor since 2019. His research interests focus on image analysis and 3-D reconstruction based on satellite/aerial multimodal data.



Han Fu received the Ph.D. degree in cartography and geographical information system from the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2021.

Since then, she worked with the State Key Laboratory of Space Information System and Integrated Application, Beijing. Her research interests focus on remote sensing image processing and analysis.



Guangjun He was born in Hubei, China, in 1987. He received the Ph.D. degree in remote sensing of resources and environment from Nanjing University, Nanjing, China, in 2015.

From 2015 to 2017, he was an Assistant Researcher Fellow with the State Key Laboratory of Space-Ground Integrated Information Technology, China Academy of Space Technology (CAST), Beijing, China, where he has been an Associate Researcher Fellow since 2018. Since 2023, he has been a Researcher with CAST. He has authored three books and over 100 articles. His research interests include intelligent interpretation of remote sensing images and multisource information fusion.



Yanming He received the B.S. and M.S. degrees in electronic information engineering from Harbin Engineering University, Harbin, China, in 2021 and 2024, respectively.

He joined China Aerospace Science and Technology Corporation, Beijing, China, where he is currently a Research Associate. His main research areas are image processing and large language models.



Zhengning Zhang received the M.Eng. degree in communication engineering from China Academy of Space Technology (CAST), Beijing, China, in 2009, and the Ph.D. degree in information and communication engineering from the Department of Electronic Engineering, Tsinghua University, Beijing, in 2022.

His research interests include remote sensing image processing, object detection, machine learning, related issues in deep learning, and remote sensing image processing.



Fengjiao Gao received the M.S. degree from Harbin University of Science and Technology, Harbin, China, in 2008.

She is currently an Associate Researcher with the Intelligent Manufacturing Research Institute, Heilongjiang Academy of Sciences, Harbin. Her research interests include remote sensing image processing and application.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:56:26 UTC from IEEE Xplore. Restrictions apply.